Entropic Dual: General Case

$$\sup_{(f,g)\in\mathscr{C}(\mathscr{X})\times\mathscr{C}(\mathscr{Y})} \int_{\mathscr{X}} f\,d\alpha + \int_{\mathscr{Y}} g\,d\beta - \varepsilon \int_{\mathscr{X}\times\mathscr{Y}} e^{\frac{-c(x,y)+f(x)+g(y)}{\varepsilon}}\,d\alpha(x)\,d\beta(y)\,.$$

- Corresponds to smoothing the constraint $f \oplus g \leq c$
- As $\varepsilon \to 0$, recovers non-entropic dual (3rd term becomes hard constraint)
- Proof that sup is attained is not trivial; cannot use c-transforms

From EOT Dual to Log-Domain